

WideCast Smart-Home Services Analysis

1. After you review the requirements specification document, comment on whether all of the listed process flows, features, and functionalities can be fully automated by the conversational AI assistant using Rasa framework/tool.

The WideCast Smart-Home Services requirements specify diverse business process workflows for managers, account specialists, technical support, and customers, covering account creation/management, order processing, incident handling, and product/service updates. Rasa's NLU customized and dialogue management capabilities support intent classification, entity extraction, can handle structured interactions, role-based access, and complex business rules also integrates with backend APIs. Most of the functionalities, like creating/updating accounts, updating placed orders, paying bills, and service incident tickets, can be fully automated using the Rasa framework which are aligned with Rasa's action server.

However, complex tasks like real-time remote control of smart home devices (lighting, thermostat, locks) may require integration with IoT systems. Additionally, few manager-level tasks (ex: deleting products) and scheduling incident tickets may need policy decisions. While Rasa can automate most of the customer facing and routine internal workflows with proper training and integration, but seamless automation of all features particularly those involving physical devices or strategic decision may require additional systems or human intervention.

2. Provide a complete list of the user stories(use-cases).

Customer User Stories

As a customer, I want to

1. Update my personal account information so I can keep my records current.
2. Pay my monthly bill so I can maintain uninterrupted services.
3. Create an order for any WideCast product or service so I can start using it.
4. Change an existing order so I can modify my service preferences.
5. Cancel an order so I can stop unwanted services.

6. Sign up for a contract plan (monthly, 1-year, or 2-year) so I can benefit from bundled offers.
7. Rent the WideCast SmartHub and necessary devices during onboarding.

Account Specialist (Customer Support) User Stories

As an account specialist, I want to

1. Create a customer account so new users can access WideCast services.
2. Update customer account information so I can maintain accurate records.
3. Create an incident ticket and assign it to a technician.
4. Place a PPV order on behalf of a customer.

Technical Support Specialist User Stories

As a technical support specialist, I want to

1. Schedule an incident ticket so I can plan support visits.
2. Close an incident ticket once the issue is resolved.
3. Cancel an incident ticket if it's no longer needed.

Manager User Stories

As a manager, I want to

1. Add new services so the company can expand offerings.
2. Update existing products or services so information stays accurate.
3. Delete obsolete products or services to keep the catalog relevant.
4. Cancel any order if needed for business or customer reasons.
5. Update any type of order to accommodate changes.
6. Delete any type of order when necessary.

System-Level or Business Rule Enforcement Stories

As a system, I must ensure

1. Customers choose at least a TV or Data plan before purchasing other services.
2. PPV services require an active Online TV plan.
3. Services like movie streaming, gaming, security, and utilities require a Data plan.
4. Apply contract-specific offers (ex: free PPV movies or premium features) based on the customer's subscription plan.

3. Provide a complete list of the entities, intents, slots, rules, responses, actions

Here's a concise list of the key components for building the WideCast Conversational AI Assistant in Rasa:

Entities:

- plan_type (TV, Data, Bundle)
- contract_duration (monthly, 1 year, 2 years)
- service_type (PPV, Movies, Games, Home Security, Utilities)
- user_role (manager, customer, support, technician)
- order_id, ticket_id, account_id
- device_type, payment_method

customer_id, technician_id, service_action (like create, update) entities also can be used.

Intents:

- create_order, update_order, cancel_order
- pay_bill, update_account, create_account
- create_incident, assign_ticket, schedule_ticket, close_ticket, cancel_ticket
- add_product, update_product, delete_product
- select_plan, inquire_offer, rent_device, greet, goodbye

instead of rent_device, control_device (like which smart device) intent is more suitable

Slots:

- user_role, plan_type, contract_duration, service_type, account_id, order_id, ticket_id, payment_status

here also customer_id, technician_id, payment_amount seems relevant, device_type

Rules:

- Require Data plan for Movies, Games, Security, Utilities
- Require TV plan for PPV services
- New users must rent Smarthub box devices
- Contract duration benefits/free offers

Authenticating user also important I believe.

Responses:

- utter_greet, utter_confirm_order, utter_payment_success, utter_error, utter_goodbye

few other utter requests would be better like utter_no_tv_plan, utter_require_service, utter_incident_status, utter_device_control. If we going for user authentication utter_auth_status

Actions:

- action_create_order, action_update_account, action_process_payment, action_apply_contract_offers, action_create_ticket, action_assign_technician

action_authenticate_user, action_cancel_order, action_update_product, action_control_device are few other actions seems relevant to add.